CareerGPS

System Analysis & Design Document

AI-Powered Career Navigation Platform

Document Type	System Analysis & Design
Version	1.0 — Initial Release
Program	Rwaad Masr Digital Scholarship
Team	12 Members (4 FE | 2 FS | 6 AI)
Architecture	Microservices + Agentic AI
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1. Problem Statement & Objectives

1.1 Problem Statement

The Egyptian and broader Arab job market faces a critical skills-to-opportunity gap. Thousands of graduates and career changers lack a clear, personalized path to bridge the gap between their current competencies and the skills demanded by employers. Existing platforms offer generic roadmaps that ignore individual backgrounds, flooding learners with undifferentiated resources and offering no accountability mechanism to sustain learning momentum.

The three root problems CareerGPS addresses are:

- Orientation Failure — Learners do not know where to start relative to their existing background, leading to wasted effort on already-known material or overwhelming advanced content.
- Resource Noise — With thousands of courses and playlists available, learners spend more time choosing what to learn than actually learning.
- Accountability Gap — Without a structured schedule and motivational feedback loop, most self-learners abandon their plans within weeks.

1.2 Project Objectives

#	Objective	Success Metric
O1	Personalized Roadmap Generation	Roadmap generated within 5 sec; user satisfaction ≥ 4/5
O2	Eliminate Resource Confusion	≥1 curated course per skill; playlist comparison available
O3	Sustain Learning Accountability	Calendar adherence rate ≥ 60% among active users
O4	Increase Job Readiness	≥70% of users report improved interview confidence
O5	Deliver MVP on Schedule	All Phase 1 features live within 4 months

2. Use Case Diagram & Descriptions

2.1 System Actors

Actor	Type	Description
Job Seeker	Primary Actor	Main user — uploads CV, generates roadmap, learns, takes exams, uses all core features
Career Explorer	Primary Actor	User who wants to explore or try a new career path via simulation before committing
Admin	Secondary Actor	Internal operator managing users, content, system health, and course catalog
Claude AI (Anthropic)	External System	Provides AI capabilities: roadmap generation, exam creation, CV building, chatbot
GitHub API	External System	Supplies public repository data and inferred technical skills for profile enrichment
LinkedIn (PDF Export)	External System	User-exported LinkedIn profile PDF, parsed for experience and skills data
Job Boards API	External System	Supplies job listings matched against user skill set (e.g. Wuzzuf, Indeed, RemoteOK)

2.2 Use Case Diagram

Use Case Diagram — CareerGPS

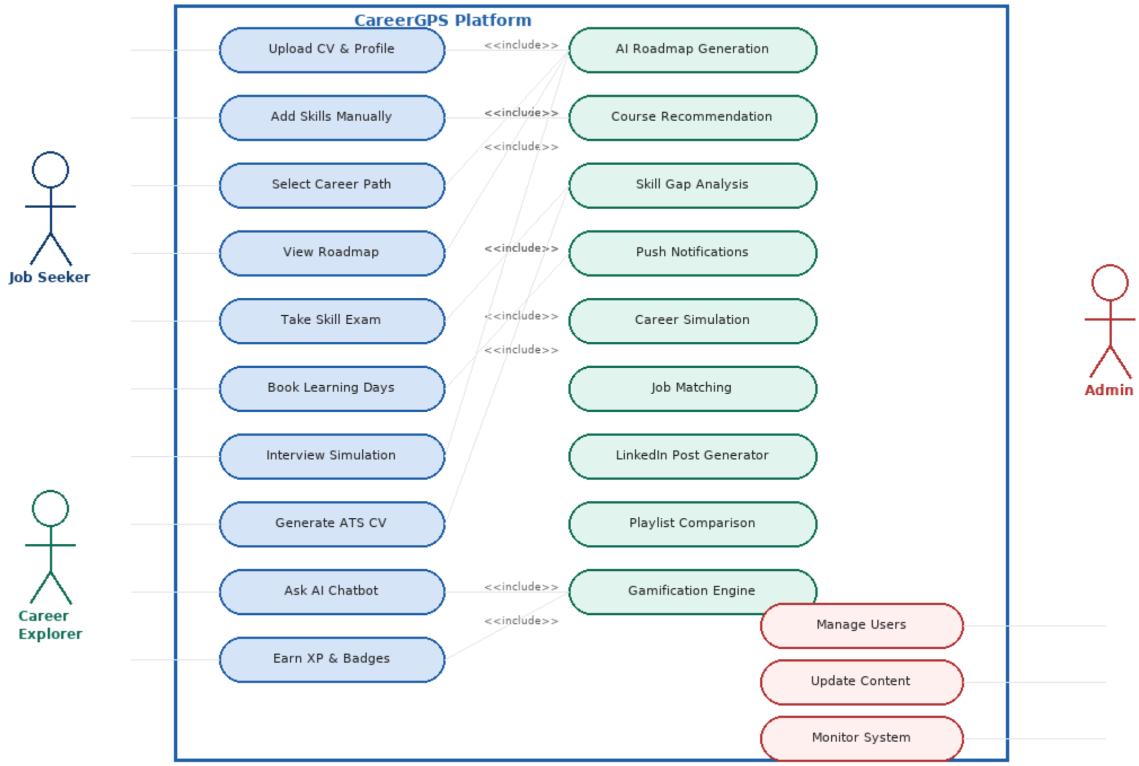

Figure 1 — CareerGPS Use Case Diagram showing all system actors and their interactions

2.3 Key Use Case Descriptions

UC-01: Upload & Parse CV

Use Case ID	UC-01
Actor	Job Seeker
Description	User uploads a PDF CV. The system parses it using AI to extract skills, experience, and education.
Precondition	User is authenticated
Postcondition	Profile is populated with extracted data
Main Flow	1. User navigates to Profile page 2. User uploads CV (PDF ≤5MB) 3. System sends CV to AI parser 4. AI extracts skills, experience, education 5. System displays extracted data for review 6. User confirms or edits the extracted profile

UC-02: Generate Personalized Roadmap

Use Case ID	UC-02

Actor	Job Seeker
Description	User selects a target career path; the AI generates a tailored roadmap with ordered skill nodes.
Precondition	Profile has skills data; User selects career
Postcondition	Roadmap saved; courses linked to each node
Main Flow	1. User selects target career from list 2. System sends profile + career to Claude AI 3. AI returns JSON roadmap (phases & skills) 4. System persists roadmap and fetches courses 5. Roadmap displayed with progress tracker

UC-03: Take Skill Exam

Use Case ID	UC-03
Actor	Job Seeker
Description	User takes an AI-generated exam for a specific skill to assess current knowledge and identify gaps.
Precondition	Skill exists in user roadmap
Postcondition	Exam results saved; gap report generated
Main Flow	1. User selects a skill and clicks 'Take Exam' 2. System generates MCQ + open-ended questions 3. User submits answers 4. AI evaluates answers 5. Gap report and resource recommendations shown

UC-04: Interview Simulation

Use Case ID	UC-04
Actor	Job Seeker
Description	User simulates a real job interview with role-specific questions and receives AI feedback.
Precondition	User has a roadmap with target career
Postcondition	Interview session saved; feedback provided
Main Flow	1. User selects Interview Simulation 2. System generates questions for career path 3. User answers each question 4. AI evaluates and scores each answer 5. Summary feedback report generated

UC-05: Generate ATS-Friendly CV

Use Case ID	UC-05
Actor	Job Seeker

Description	User provides a job description; the AI generates a tailored CV with ATS keywords.
Precondition	User has a completed profile
Postcondition	New CV PDF generated and downloadable
Main Flow	1. User pastes or uploads job description 2. System extracts keywords from JD 3. AI generates CV matching user profile + JD keywords 4. CV rendered in template 5. User downloads CV as PDF

3. Functional & Non-Functional Requirements Summary

3.1 Functional Requirements (Condensed)

Full functional requirements are detailed in the Requirements Specification Document (v1.0). Key capabilities by module:

Module	Core Capabilities
Profile	CV parsing (AI), LinkedIn PDF import, GitHub API integration, manual skill management
Roadmap	AI-generated personalized roadmap, phase-based skill ordering, progress tracking, roadmap regeneration
Courses	Per-skill course recommendation, YouTube playlist comparison, external course linking
Calendar	Study schedule builder, day/time selection, persistent reminder configuration
Assessments	AI-generated skill exams (MCQ + open-ended), gap analysis, resource recommendations per gap
CV Builder	ATS-keyword extraction from JD, tailored CV generation, from-scratch CV creation, PDF export
Interview	Role-specific mock interview, AI answer evaluation, scoring, feedback report
Gamification	XP point system, badge library, level progression, leaderboard
Career Sim	Hands-on guided career simulation tasks (TryHackMe-style), multiple careers supported
Notifications	Push notifications (FCM), AI-generated motivational messages, email reminders
Jobs	Skill-based job matching from external recruitment APIs, relevance scoring
Admin	User management, content moderation, platform analytics, system health monitoring

3.2 Non-Functional Requirements

Category	Requirement
Performance	Roadmap generation < 5 sec; CV parsing < 10 sec at 95th percentile
Scalability	Support 10,000 concurrent users; horizontal scaling via container orchestration (Docker + K8s)
Availability	99.5% uptime SLA; auto-restart on service failure; health check endpoints per microservice
Security	AES-256 at rest; TLS 1.3 in transit; JWT + refresh token rotation; OWASP Top 10 compliance
Usability	Core task (upload CV → view roadmap) achievable in ≤3 steps; WCAG 2.1 Level AA
Localization	Full support for English (LTR) and Arabic (RTL)

Category	Requirement
Maintainability	Microservices architecture; each service independently deployable; 80%+ test coverage target
Privacy	GDPR-compliant; user can request data export or deletion at any time

4. Software Architecture

4.1 Architecture Style

CareerGPS adopts a Microservices Architecture combined with an Agentic AI Layer. Each domain feature is implemented as an independently deployable service, communicating via REST APIs and an async message queue. This design ensures:

- Independent scaling of AI-heavy services (roadmap generation, exam creation) without affecting UI responsiveness.
- Fault isolation — a failing Course Service does not bring down the Roadmap Service.
- Team parallelism — the 4 Frontend, 2 Full-Stack, and 6 AI team members can develop and deploy independently.
- Future extensibility — adding a new feature (e.g., Freelance Matching) requires only a new microservice, not modifying existing ones.

4.2 Architecture Diagram

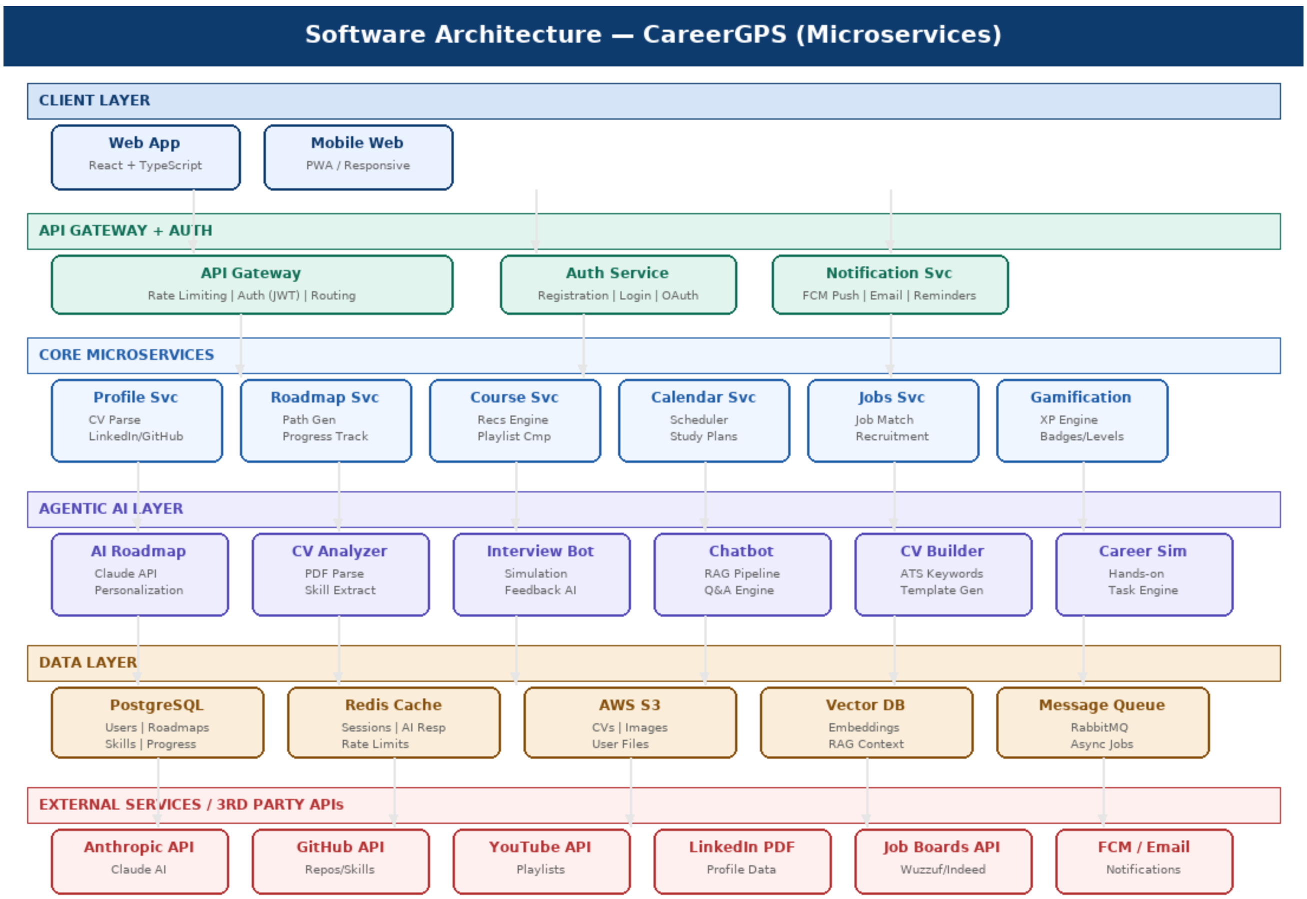

Figure 2 — CareerGPS Software Architecture (6-layer Microservices + Agentic AI)

4.3 Component Descriptions

Component	Technology	Responsibility
Client Layer	React + TypeScript PWA	Renders UI; communicates with API Gateway via REST/WebSocket
API Gateway	Node.js + Express	Routes requests, enforces auth (JWT), rate-limiting, load balancing
Auth Service	Clerk / custom JWT	Handles registration, login, OAuth with LinkedIn/GitHub
Profile Service	Node.js microservice	CV upload, AI-based parsing, GitHub integration, skill management
Roadmap Service	Node.js microservice	Roadmap generation, node tracking, progress updates
Course Service	Node.js microservice	Course recommendations, playlist comparison engine
AI Agent Layer	Python (LangChain/LangGraph)	Orchestrates calls to Claude API for roadmap, exams, CV builder, chatbot
Gamification Service	Node.js microservice	XP calculation, badge awards, leaderboard management
Notification Service	FCM + SendGrid	Push notifications, motivational messages, email reminders
Database	PostgreSQL + Redis	Persistent storage + caching for sessions and AI responses
File Storage	AWS S3 / Cloudflare R2	Stores uploaded CVs, profile photos, generated documents
Message Queue	RabbitMQ	Async job processing for AI tasks, notification dispatch
Vector Database	Pinecone / Chroma	Stores embeddings for RAG pipeline used by chatbot and recommendations

5. Data Flow Diagram (DFD)

5.1 Level 0 — Context Diagram

The context diagram shows the system as a single process and identifies all external entities that interact with it, along with the data flows between them.

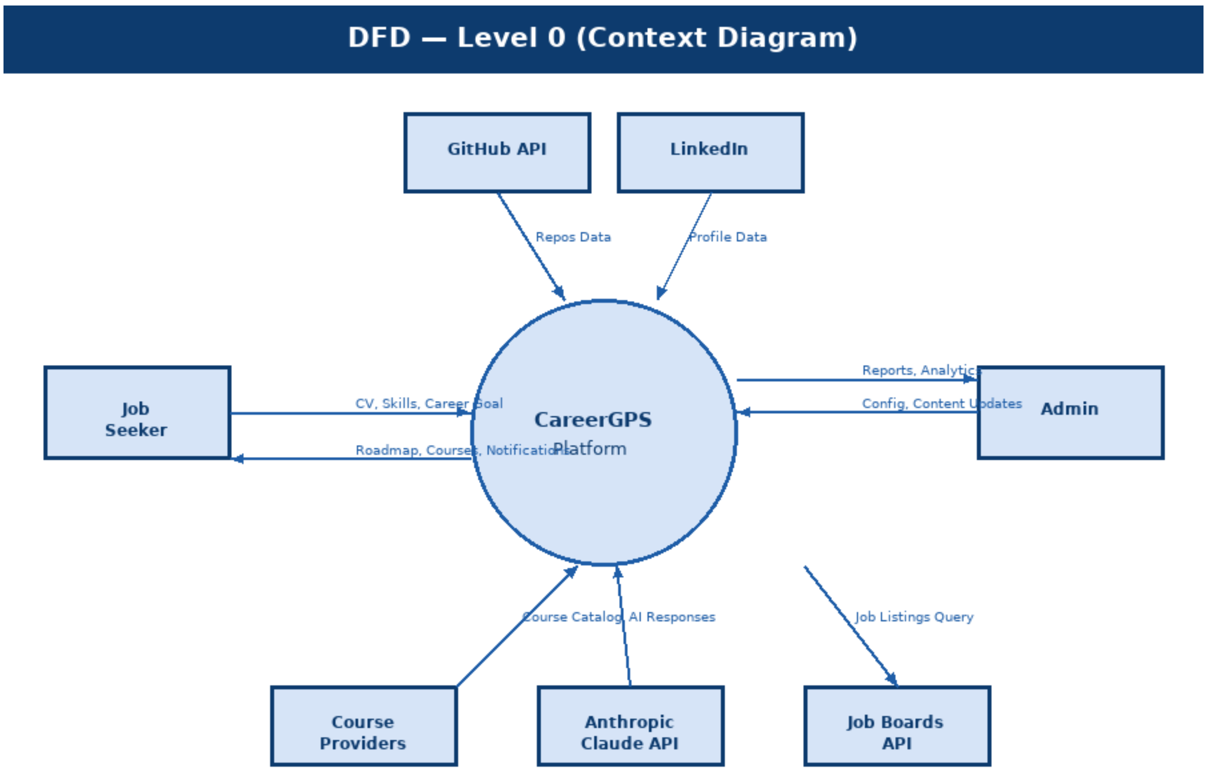

Figure 3 — Level 0 Context DFD showing all external entities and high-level data flows

5.2 Key Data Flows

Source	Destination	Data Flow Description
Job Seeker	System	CV file (PDF), career selection, exam answers, schedule preferences
System	Job Seeker	Personalized roadmap, course list, exam results, notifications, generated CV
GitHub API	System	Repository list, languages used, commit activity, inferred technical skills
LinkedIn PDF	System	Parsed experience, education, current role, connections data
Anthropic Claude	System	Structured roadmap JSON, exam questions, interview feedback, CV content
System	Anthropic Claude	User profile data, career goal, skill list, exam answers, job description

Source	Destination	Data Flow Description
Course Providers	System	Course metadata (title, URL, rating, provider, duration)
Job Boards API	System	Job listings with required skills, location, salary range
Admin	System	Content updates, user actions, configuration changes
System	Admin	Analytics reports, user statistics, error logs, system metrics

6. Sequence Diagram

6.1 Roadmap Generation Flow

This sequence diagram illustrates the end-to-end interaction between system components when a user uploads their CV and requests a personalized roadmap. It covers CV parsing, AI roadmap generation, course linking, and database persistence.

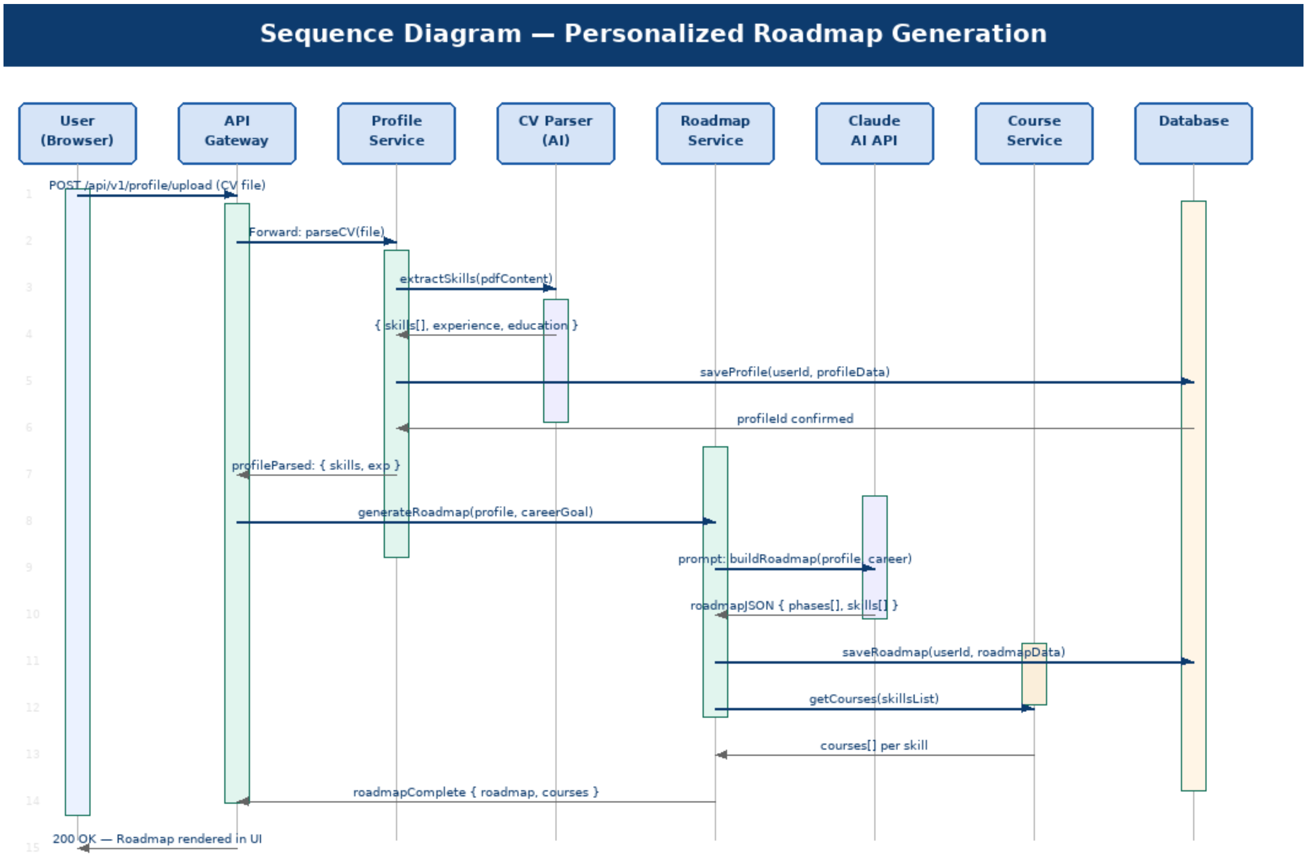

Figure 4 — Sequence Diagram for Roadmap Generation (CV Upload → AI Processing → Roadmap Display)

6.2 Sequence Step Summary

Step	From	To	Action
1	User (Browser)	API Gateway	POST /api/v1/profile/upload with CV file
2	API Gateway	Profile Service	Forward authenticated request with file
3	Profile Service	CV Parser (AI)	extractSkills(pdfContent) — calls AI extraction
4	CV Parser	Profile Service	Returns { skills[], experience, education }
5	Profile Service	Database	Persist profile data linked to userId
6	Profile Service	API Gateway	Return profileParsed confirmation
7	API Gateway	Roadmap Service	generateRoadmap(profile, careerGoal)

Step	From	To	Action
8	Roadmap Service	Claude AI API	Send structured prompt with user profile + career
9	Claude AI API	Roadmap Service	Return roadmapJSON { phases[], skills[] }
10	Roadmap Service	Course Service	getCourses(skillsList) for each roadmap node
11	Roadmap Service	Database	Persist complete roadmap with course links
12	API Gateway	User	200 OK — Complete roadmap JSON rendered in UI

7. Activity Diagram — User Onboarding Flow

The activity diagram below models the complete user journey from first registration through roadmap generation and active learning, including decision points and parallel flows.

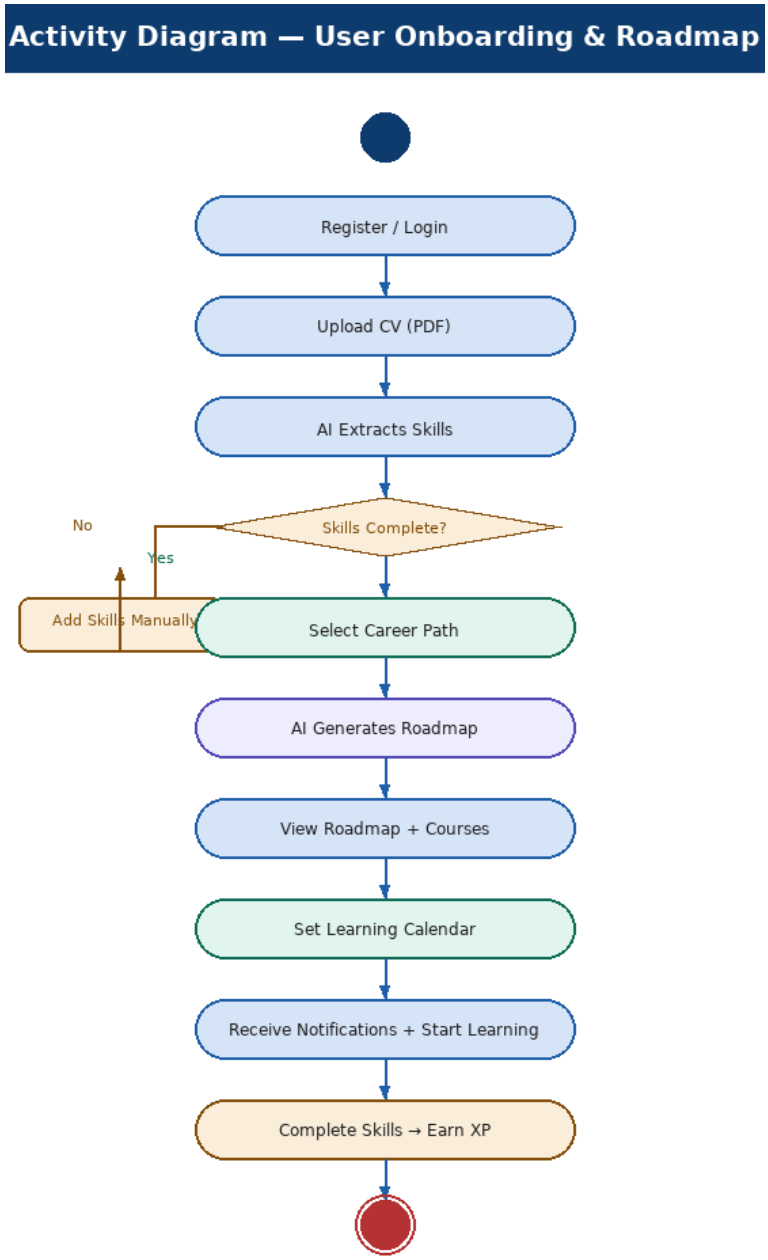

Figure 5 — Activity Diagram showing the end-to-end user onboarding and learning workflow

8. State Diagram — Skill Learning Lifecycle

Each skill node in a user's roadmap transitions through a defined set of states. The state diagram below captures all possible states and the triggers that drive transitions between them.

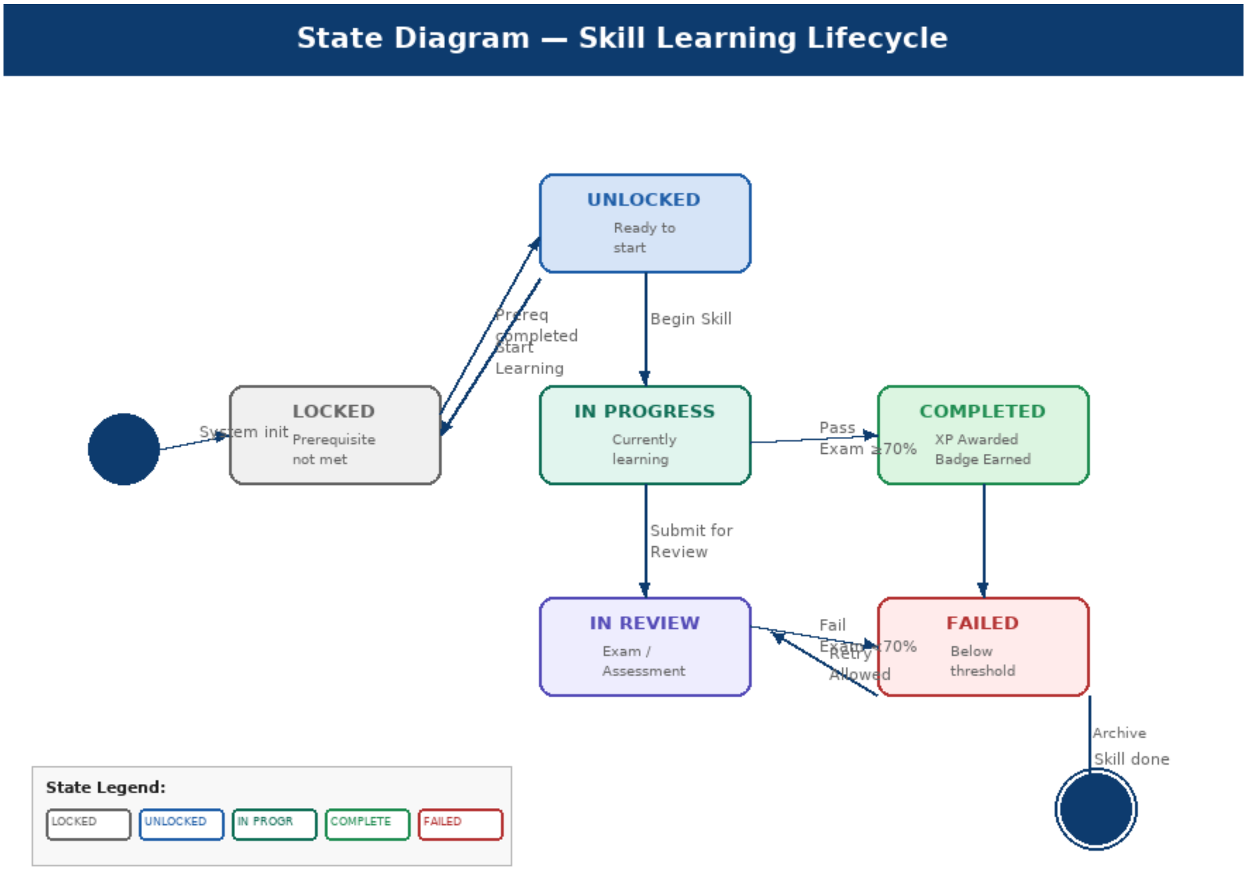

Figure 6 — State Diagram for a Roadmap Skill Node

State	Description	Transition Trigger
LOCKED	Skill is not yet accessible; prerequisites not met	Previous skill node is marked COMPLETED
UNLOCKED	Skill is accessible and ready to start	User explicitly clicks 'Start Skill'
IN PROGRESS	User is actively learning the skill	User begins course or study session
IN REVIEW	User submitted skill for assessment (exam)	User submits skill exam answers
COMPLETED	Skill mastered; XP and badge awarded	Exam score ≥ 70% pass mark
FAILED	Exam score below threshold	Exam score < 70%; retry allowed after 24h

9. Class Diagram — Core Domain Model

The class diagram defines the key domain entities of CareerGPS, their attributes, core methods, and relationships. This serves as the blueprint for both the database schema and the API data models.

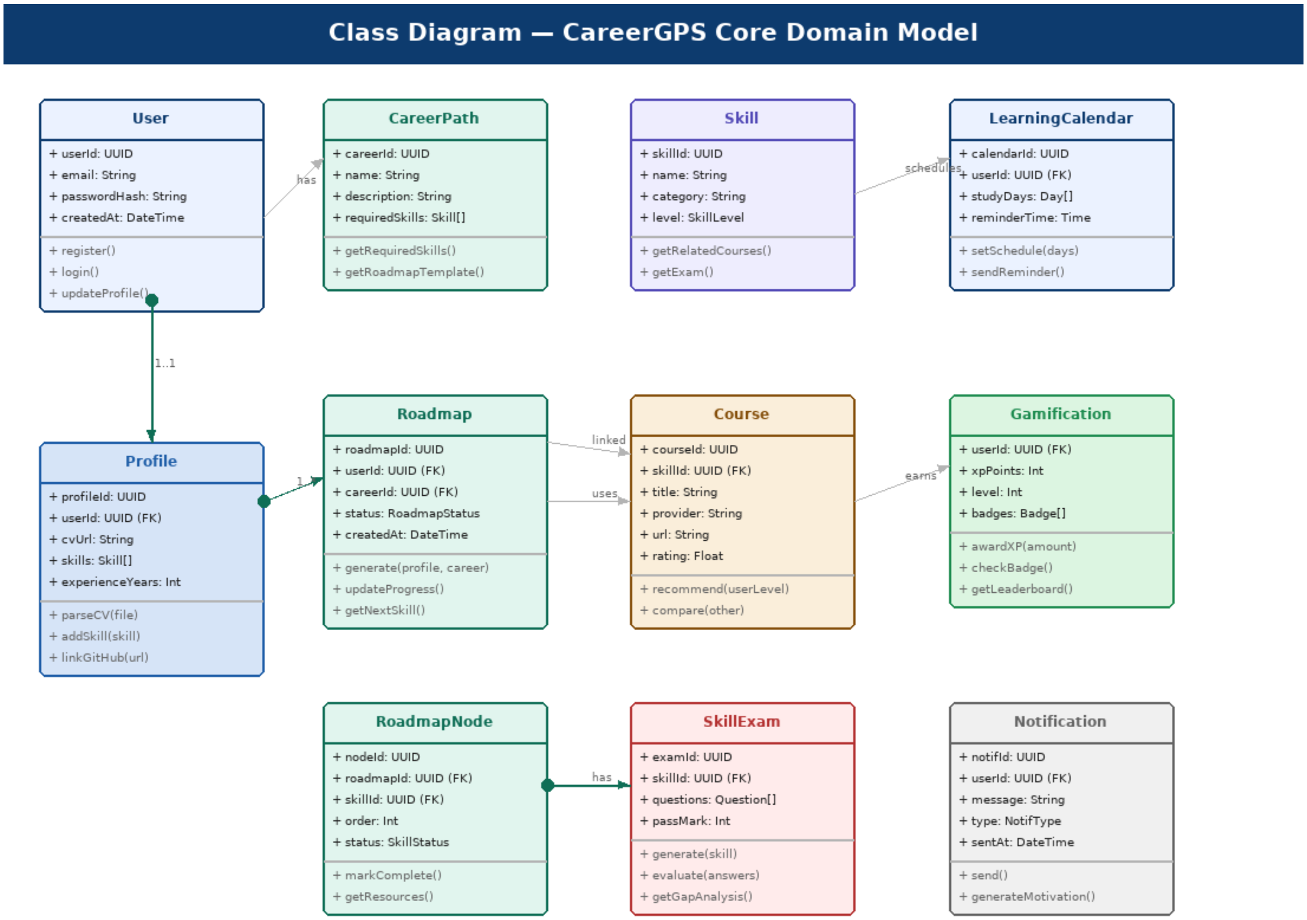

Figure 7 — CareerGPS Class Diagram showing core domain model and entity relationships

9.1 Key Relationships

Class A	Class B	Relationship	Description
User	Profile	1:1 Composition	Each user has exactly one profile; profile cannot exist without user
User	Roadmap	1:N Composition	A user can have multiple roadmaps (different career paths)
Roadmap	RoadmapNode	1:N Composition	Each roadmap contains an ordered list of skill nodes
RoadmapNode	Skill	N:1 Association	Each roadmap node references a skill from the global skill catalog

Class A	Class B	Relationship	Description
Skill	Course	1:N Association	Each skill can have multiple recommended courses
Skill	SkillExam	1:1 Association	Each skill has one generated exam for assessment
User	Gamification	1:1 Association	Each user has a gamification profile tracking XP, level, badges
User	LearningCalendar	1:1 Association	Each user manages one learning calendar with study schedule
User	CareerPath	N:M Association	Users can explore multiple career paths; each path has many users

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